



ALISHA GULHANE

AI & Data Science Undergraduate | Aspiring ML & AI Engineer

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• Pune, Maharashtra, India

PROFILE

Motivated AI & Data Science undergraduate with hands-on experience in machine learning, data analytics, and full-stack development. Skilled in building intelligent, scalable solutions through projects in predictive modelling, real-time systems, database design, and web development. Recognised patent holder for an **Automated Image-Based System for Identifying Lost and Found Items**. Passionate about applying AI to solve real-world problems, with growing expertise in Machine Learning, Deep Learning, and Generative AI.

PATENT



Automated Image-Based System for Identifying Lost and Found Items

Patent Holder • Developed an AI-powered system leveraging computer vision and image recognition to automatically match lost items with found item reports, streamlining the recovery process through intelligent image comparison and classification.

TECHNICAL SKILLS

Languages	Python · C · SQL · JavaScript
AI / ML	Scikit-learn · Pandas · NumPy · Regression Models · Decision Trees · Feature Engineering · EDA · Model Evaluation , RAG
Deep Learning	Deep Learning Fundamentals · Anomaly Detection · Generative AI · Diffusion Models
Databases	MySQL · Database Design · SQL Queries · Data Storage & Retrieval
Web Dev	HTML · CSS · JavaScript · Frontend Development · User Authentication
Analytics	Statistical Analysis · Matplotlib · Data Visualisation · Report Generation
Tools	Jupyter Notebook · VS Code · GitHub · Anaconda

CORE PROJECTS

Lost and Found Portal

Frontend & Database Developer [Patent Holder]

- Developed a responsive web portal for users to report and recover lost items using HTML, CSS, and JavaScript.
- Implemented backend logic and database functionality to enable lost-found entry matching.
- Patent granted for the **Automated Image-Based System for Identifying Lost and Found Items** — AI-driven image recognition to automate item matching.

SmartAgri: AI-Based Crop Health & Yield Prediction Platform

Frontend & Database Developer, AI/ML Engineer & Research Co-Author

- Developed a full-stack AI-powered precision agriculture platform integrating **crop recommendation**, **plant disease detection**, and **crop yield prediction** into a unified advisory system.
- Built an ensemble deep learning framework using **EfficientNet-B0**, **EfficientNet-B1**, and **ResNet-50**, achieving **97.1% disease classification accuracy** across 52 crop disease classes.
- Implemented **XGBoost-based yield prediction** with engineered seasonal and climatic features, achieving **$R^2 = 0.83$** on agricultural datasets.
- Designed explainable AI functionality using **Grad-CAM++** for disease localization and prediction transparency.
- Developed multilingual advisory support (**English, Hindi, Marathi**) through a **React + FastAPI** architecture.
- Co-authored an **IEEE-format research paper** titled "*SmartAgri: An Ensemble-Based Multi-Task Artificial Intelligence Platform for Crop Disease Detection, Yield Prediction and Decision Support in Precision Agriculture.*"

Smart Job Portal

Full Stack Developer & Analyst

- Developed a job search and application portal for students and recruiters using Python and MySQL.
- Designed database schemas to manage job listings, applicant profiles, and secure authentication systems.
- Analysed user data to improve job matching accuracy and personalised recommendations.

Intelligent Personal Protective Headgear (IPPH)

ML & Systems Lead[hardware and software]

- Built a smart safety helmet using sensors and AI to detect accidents and hazardous conditions in real time.
- Designed a data pipeline for sensor data collection, preprocessing, and automated alert generation.
- Implemented predictive logic to trigger emergency notifications and location tracking for safety response.

EXPERIENCE & LEADERSHIP

Event Coordinator | VEDC, Pune

January 2025 – Present

- Coordinate entrepreneurship events, startup activities, and participant logistics for student-led innovation programmes.
- Manage event data and improve team coordination through structured planning and communication workflows.

Cultural Head | CMIS, Pune

June 2020 – April 2021

EDUCATION

B.Tech in Artificial Intelligence & Data Science

Vishwakarma Institute of Technology (VIT), Pune

Expected: May 2028

CERTIFICATIONS

Generative AI with Diffusion Models — NVIDIA

Mar 2026

Applications of AI for Anomaly Detection — NVIDIA

Mar 2026

Fundamentals of Deep Learning — NVIDIA

Feb 2026

People Management

Dec 2025

Artificial Intelligence Bootcamp — C-DAC Pune & VIT (FutureSkills PRIME)

Aug 2025

Data Analysis (Excel) — Infosys Springboard

2024

Python Programming — Infosys Springboard

2024

C Programming — Infosys Springboard

2024

Database Management Systems (DBMS) — Infosys Springboard

2024

LANGUAGES

English (Advanced) • Hindi (Professional) • Marathi (Advanced)

HOBBIES

Research writing, Problem Solving, Artificial Intelligence & Machine Learning , Computer Vision , Explainable AI (XAI) , Precision Agriculture Technologies , Full-Stack Development , Entrepreneurship & Innovation, Public Speaking.